

Certified bounds for the least Carmichael numbers with 64–144 prime factors

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Abstract

Let S_k denote the smallest Carmichael number with exactly k distinct prime factors (OEIS [A006931](#); known for $k \leq 63$, open beyond). We give certified lower and upper bounds on S_k for every k from 64 to 144, each with a reproducible certificate and exact endpoints, and quantify the improvement over the identified previous bound. No claim of priority is made; the bounds stand on the computations. The sharpest cases are $k = 64$ and $k = 65$. For $k = 64$ the upper bound is a 148–digit Carmichael number N with exactly 64 prime factors found by J. Webster (communicated September 11, 2026) and verified here, so $S_{64} \leq N$; our own 148–digit incumbent N_0 , found by a GPU exchange search, is approximately $1.03 N$ and is shown to be the least 64–factor Carmichael number whose Carmichael function divides $\lambda(N_0)$. For the lower bound we prove, by a finite exhaustion over a certified prime universe, that no 64–factor Carmichael number is smaller than 10^{145} . Together,

$$10^{145} \leq S_{64} \leq N < 10^{148},$$

so S_{64} has 146, 147, or 148 decimal digits. All arithmetic is exact. The lower bound is a reproducible computation: the admissible universe is bounded by a Korselt cofactor argument, the resulting bound is recomputed independently, and the exhaustion is carried out by three separate enumeration implementations, one of which takes its Korselt verdict at every leaf from a frozen reference oracle. The same cofactor minimum transfers to the next factor count, giving $10^{148} \leq S_{65} \leq U_{65}$ for an explicit verified 151–digit Carmichael number U_{65} , so the smallest 65–factor Carmichael number has 149, 150, or 151 digits (§5). For every k up to 144 the same relaxed cofactor bound, computed in one shared pass, gives a certified lower endpoint that improves the elementary primorial baseline by 16–29 digits, paired with a verified incumbent as upper endpoint (§6). The author is responsible for the mathematics and code; see the acknowledgements for the role of AI assistance.

1 Statement

A *Carmichael number* is a composite integer n with $b^n \equiv b \pmod{n}$ for all integers b . By Korselt’s criterion, a composite integer n is Carmichael if and only if it is squarefree and $(p-1) \mid (n-1)$ for every prime $p \mid n$. Write $\omega(n)$ for the number of distinct prime factors. The sequence $S_k = \min\{n : n \text{ Carmichael}, \omega(n) = k\}$ is OEIS [A006931](#); values are known for $3 \leq k \leq 63$ (R. G. E. Pinch for the smaller k ; the $k = 36$ – 63 values from the Butler University computation, collected in J. Webster’s repository cited below), and S_{64} is open as of September 2026. This note proves:

Theorem 1 (Interval). $10^{145} \leq S_{64} \leq N$, where

$$\begin{aligned} N = & 524394558743645786915728373569633336575821258 \\ & 922388902572826761352267639397817207104051794 \\ & 486290626375080033731873206687812206750535842 \\ & 2938650976001. \end{aligned}$$

is a 148-digit Carmichael number with exactly 64 prime factors. Consequently S_{64} has 146, 147, or 148 decimal digits, and the remaining exclusion interval for S_{64} is $[10^{145}, N)$.

The two bounds are proved in §2 and §3. We use throughout the following elementary consequence of Korselt’s criterion.

Lemma 1 (Pairwise admissibility). *If $p < q$ are distinct primes dividing a Carmichael number n , then $p \nmid (q - 1)$. Moreover n is odd.*

Proof. Since $q \mid n$ we have $(q - 1) \mid (n - 1)$, so $n \equiv 1 \pmod{d}$ for every divisor d of $q - 1$. If $p \mid (q - 1)$ then $n \equiv 1 \pmod{p}$; but $p \mid n$ gives $n \equiv 0 \pmod{p}$, forcing $p \mid 1$, a contradiction. Oddness: if n were even then $n - 1$ would be odd, so for any odd prime factor p the even number $p - 1$ could not divide $n - 1$; hence a Carmichael number with $k \geq 3$ distinct prime factors has no factor 2 and is odd. $\square$

2 Upper bound

The number N of Theorem 1 is the product of the 64 primes

19, 29, 31, 37, 41, 43, 47, 53, 61, 67, 71, 73, 79, 89, 97, 101, 103, 109, 113, 127,
131, 139, 151, 157, 163, 167, 181, 193, 197, 199, 211, 239, 241, 251, 257, 271, 277, 281, 307, 313,
331, 337, 353, 379, 397, 409, 421, 433, 443, 449, 463, 499, 521, 547, 577, 599, 601, 617, 631, 641,
691, 769, 1249, 1579.

These are 64 distinct primes; re-derive N by multiplying the list rather than trusting the printed decimal. Each factor is at most 1579, so its primality is certified by exact trial division (a deterministic check at this size). One verifies directly that $(p - 1) \mid (N - 1)$ for every p ; equivalently, writing $\lambda(N) = \text{lcm}_{p \mid N}(p - 1) = 155016423578016000$, one has $N \equiv 1 \pmod{\lambda(N)}$. Hence N is a squarefree Carmichael number with $\omega(N) = 64$, and $S_{64} \leq N$. Since $10^{147} \leq N < 10^{148}$, the number N has 148 digits. The certificate is verified by `ref/ref_carmichael.py::verify_certificate`, an independent, dependency-free routine (squarefreeness, distinctness, per-factor primality, and Korselt) sharing no code with the search that produced N .

Remark 1 (Provenance). N was found by J. Webster’s CPU search and communicated to the author on September 11, 2026; it is reproduced here after independent factorisation and verification. It is an upper bound, not claimed minimal. Our own best incumbent, found by the GPU exchange search of §2.1 at radius $r = 8$ and modulus $M_0 = 1768248177696000$, is the 148-digit number

$$N_0 \approx 1.0295 N \quad (N_0/N = 1.029452793\dots), \quad \lambda(N_0) = M_0, \quad P^+(N_0) = 1249,$$

whose 64 factors are listed in the repository file `results_k64_best_global_prev_148_539839443357.json`; N and N_0 share 53 prime factors. N_0

had improved our previous 148-digit incumbent (largest prime 3697, radius 7, same modulus) by a factor 1.605 and an earlier 149-digit candidate. The 148 digits of N are consistent with the known neighbours only in the following precise sense: Webster's N_{63} has 145 digits, hence $N_{63} < 10^{145} \leq S_{64}$, and the N_{65} candidate has 151 digits, hence exceeds N . This uses the proved interval and the stated digit lengths; it is not an appeal to any general monotonicity of S_k .

2.1 The exchange search and a restricted minimum

Fix a modulus M and let $Q(M) = \{q \text{ prime} : (q-1) \mid M, q \nmid M\}$. If n is composite and squarefree with every prime factor in $Q(M)$ and $n \equiv 1 \pmod{M}$, then $(q-1) \mid M \mid (n-1)$ for every $q \mid n$, so n is Carmichael by Korselt. The converse holds only for the exact modulus: every Carmichael number n with $\lambda(n) = D$ has all its prime factors in $Q(D)$ (a prime factor q of a Carmichael number never divides $\lambda(n)$, by Lemma 1) and satisfies $n \equiv 1 \pmod{D}$. Consequently searching every divisor D of M covers every Carmichael number with $\lambda(n) \mid M$, whereas a search at M alone does not (for instance $\lambda(561) = 80 \mid 160$ but $561 \not\equiv 1 \pmod{160}$).

Exchange meet-in-the-middle. Choose a base $B_0 \subset Q(M)$ of 64 primes with product P_B , and a radius r . A *radius- r exchange* replaces $D \subset B_0$ by $I \subset Q(M) \setminus B_0$ with $|D| = |I| = r$, giving $n = P_B P_I / P_D$ where $P_I = \prod I$, $P_D = \prod D$. The condition $n \equiv 1 \pmod{M}$ reads $P_B P_I \equiv P_D \pmod{M}$, so the search is a join of the multiset $\{P_D \bmod M\}$ over deletions with $\{P_B P_I \bmod M\}$ over insertions. In the implementation the deletion side is generated on the CPU (with the product filter below) and held resident on the GPU, the insertion side is streamed from a CPU generator and sorted and joined on the GPU (very small instances, and the 134-bit case at radius 14, are joined on the host with exact integers); every residue match is checked with exact integer arithmetic, and every candidate below the bound is verified by the reference oracle. An *improving* completion is one with $n < U$ for the current bound U .

Lemma 2 (Product bounds). *Let U be a positive integer bound, $r \geq 1$, let $D_{\max,r}$ be the product of the r largest primes of B_0 and $I_{\min,r}$ the product of the r smallest primes of $Q(M) \setminus B_0$ (empty products are 1). Every improving radius- r completion satisfies*

$$P_I < I_{\text{cap}} := \left\lfloor \frac{U D_{\max,r}}{P_B} \right\rfloor + 1 \quad \text{and} \quad P_D \geq D_{\min} := \left\lfloor \frac{P_B I_{\min,r}}{U} \right\rfloor + 1,$$

the largest prime of I is at most $\lfloor (I_{\text{cap}} - 1) / I_{\min,r-1} \rfloor$, and if the insertion primes are split at a threshold y into $a_1 < a_2 < \dots \leq y < c_1 < c_2 < \dots$, a completion using exactly h primes above y satisfies $n \geq (P_B / D_{\max,r}) \prod_{j \leq r-h} a_j \prod_{j \leq h} c_j$.

Proof. $n < U$ with $P_D \leq D_{\max,r}$ gives $P_I < U D_{\max,r} / P_B$, hence $P_I \leq \lfloor U D_{\max,r} / P_B \rfloor$; with $P_I \geq I_{\min,r}$ it gives $P_D > P_B I_{\min,r} / U$. The largest prime of I times the product of the other $r-1$ insertion primes, which is at least $I_{\min,r-1}$, is $P_I \leq I_{\text{cap}} - 1$; the stratified bound is the stratified minimum of P_I . (A completion admitted by the integer rounding but not improving is discarded by the final test $n < U$.) $\square$

The second bound is what makes large radii affordable: at $r = 9$ for N_0 it keeps 40 966 814 of the $\binom{64}{9} = 27\,540\,584\,512$ deletion subsets, so the deletion side stays resident on the GPU while the insertion side is streamed through it.

Lemma 3 (Per-modulus exactness). *Let $D \mid M$ with $|Q(D)| \geq 64$, let $B_0(D)$ be the 64 smallest primes of $Q(D)$, T_{64} their product and $T_{63} = T_{64} / \max B_0(D)$, and suppose $T_{64} < U$. Let $S \subset Q(D)$*

be any 64-subset with $\prod S < U$. Then every prime of S is at most $\lfloor (U-1)/T_{63} \rfloor$, and $|S \setminus B_0(D)| \leq r_{\max}(D)$, where, with $a_1 < a_2 < \dots$ the primes of $Q(D) \setminus B_0(D)$, $b_1 > b_2 > \dots > b_{64}$ the primes of $B_0(D)$ in decreasing order, $R(D) := \min(64, |Q(D)| - 64)$, $L_0 := T_{64}$ and $L_r := T_{64} \prod_{j \leq r} a_j / \prod_{j \leq r} b_j$ for $1 \leq r \leq R(D)$,

$$r_{\max}(D) := \max\{r \in \mathbb{Z} : 0 \leq r \leq R(D), L_r < U\}$$

(well defined, since $L_0 = T_{64} < U$). In particular the exchange search from $B_0(D)$ at radii $0, \dots, r_{\max}(D)$ over the pool $\{q \in Q(D) : q \leq (U-1)/T_{63}\}$ finds every Carmichael number $n < U$ with $\omega(n) = 64$ and $\lambda(n) = D$.

Proof. The 63 smallest elements of S multiply to at least T_{63} , which bounds the largest by $(U-1)/T_{63}$. If $|S \setminus B_0(D)| = r$ then $r \leq \min(64, |Q(D)| - 64)$ and $\prod S \geq L_r$, because the r primes outside are at least $a_1, \dots, a_r$ and the $64 - r$ primes inside are at least the $64 - r$ smallest of $B_0(D)$; and whenever $r + 1$ is in range, $L_{r+1}/L_r = a_{r+1}/b_{r+1} > 1$ because every prime outside $B_0(D)$ exceeds every prime inside. Hence $r > r_{\max}(D)$ forces $\prod S \geq L_r \geq L_{r_{\max}(D)+1} \geq U$. The last sentence follows from the first paragraph of this subsection. $\square$

Theorem 2 (A restricted minimum). *N_0 is the least Carmichael number with exactly 64 prime factors whose Carmichael function divides $M_0 = \lambda(N_0) = 2^8 \cdot 3^5 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 83$. Consequently every 64-factor Carmichael number below N_0 —in particular N —has a prime factor q with $(q-1) \nmid M_0$.*

Proof. If $\lambda(n) \mid M_0$ then $\lambda(n) = D$ for a divisor D of M_0 . Of the 20 736 divisors, 111 satisfy the hypothesis $T_{64} < N_0$ of Lemma 3 (the others cannot carry such an n at all); their pool caps and radii $r_{\max}(D) \leq 14$ are tabulated in `gpu/mstar_divisor_family.json`. For each of the 111 moduli the exchange search of Lemma 3 was run in full—base $B_0(D)$ checked directly, radii $1, \dots, r_{\max}(D)$ over the complete capped pool with the bounds of Lemma 2, and the residue join $n \equiv 1 \pmod{D}$ —for 684 exchange instances in all, with no incomplete instance and no completion below N_0 . The deletion and insertion record counts at $D = M_0$ (for instance 36 755 836 and 652 881 611 at radius 9, and 13 and 23 at radius 14) agree exactly with an independent enumeration. $\square$

Webster’s N illustrates the corollary: exactly one of its factors, 1579 ($1578 = 2 \cdot 3 \cdot 263$), has $(q-1) \nmid M_0$, and $\lambda(N) = M_0 \cdot 263/3$. The same census and search were then run for N itself. $\lambda(N) = 2^8 \cdot 3^4 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 83 \cdot 263$ has 34 560 divisors, of which 118 satisfy $T_{64} < N$ (radii $r_{\max}(D) \leq 13$; table `gpu/divisor_family_lambdaJ.json`); all 118 were searched in full (728 exchange instances plus 118 base checks, none incomplete) with no completion below N . Hence:

Theorem 3 (A restricted minimum for N). *N is the least Carmichael number with exactly 64 prime factors whose Carmichael function divides $\lambda(N)$; every 64-factor Carmichael number below N has a prime factor q with $(q-1) \nmid \lambda(N)$ and a prime factor q' , not necessarily distinct from q , with $(q'-1) \nmid M_0$ (the latter by Theorem 2, since $n < N < N_0$).*

Both restricted minima are statements about a divisor family, not about S_{64} . The divisor-family computations reported here do not cover all possible Carmichael functions below N and therefore do not certify global minimality: every competitor $n < N$ has $\lambda(n)$ dividing $L = \text{lcm}\{q-1 : q \leq (N-1)/C_{63} \text{ prime}\}$, a finite but astronomically large family of moduli (more than $10^{10\,000}$ divisors of L pass the pool-size and cheapest-product tests), and no argument is offered here that a modulus-first enumeration competes with preproduct exhaustion. Webster’s separate exhaustive search is designed to provide global coverage by enumerating preproducts P and computing $\lambda(P)$ from their prime factors, following the preproduct/composite-completion approach of Shallue and Webster (*Algorithms for Carmichael numbers*, §5 and §7.4); the status of his $k = 64$ computation is

as communicated by him. The census script (`gpu/divisor_family_census.py`) is standard-library Python and independent of the GPU code; it discards a divisor with fewer than 64 eligible primes below its sieve bound only after checking that no completion with primes beyond the bound could be below U .

Records and reproduction. An implementation error found in review (the insertion cap excluded the single largest affordable insertion product; in these two family computations this affected only radius 1) was corrected, and both families were then re-run in full with the corrected driver; the records cited here are those re-runs. The 272 completions omitted by the original runs were also checked directly, with no residue match. The runner uses one frozen bound for the census, the base checks and the exchange instances, has no default time limit, and exits with a nonzero status unless every requested modulus completed every requested radius; an independent coverage check re-reads the records. The per-modulus records are in the repository under `gpu/records/M0_U539839443357/D<D>.log` (family of M_0 , bound N_0) and `gpu/records/lambdaN_U524394558743/D<D>.log` (family of $\lambda(N)$, bound N); the census tables are `gpu/mstar_divisor_family.json` and `gpu/divisor_family_lambdaJ.json`. To reproduce (the search needs a CUDA host with the two dumpers built; `\` marks a shell continuation):

```
python3 gpu/divisor_family_census.py <incumbent.json> <census.json>
FAM_TAG=<tag> FAMILY_JSON=<census.json> U_OVERRIDE=<bound> \
python3 gpu/run_mstar_family.py
python3 gpu/verify_family_coverage.py <census.json> \
gpu/records/<tag>_U<bound prefix> <bound>
```

The pipeline's positive control, which rediscovers N_0 from the 64 smallest primes of $Q(M_0)$ at radius 8 with the bound frozen at the previous incumbent, is

```
MODULUS=1768248177696000 U_OVERRIDE=<previous incumbent> DFILTER=1 CKPT_TAG=control \
R_MIN=8 R_MAX=8 BASES=ksmall INS_PRIME_CAP=4000 python3 gpu/mitm_gpu_mstar_driver.py
```

The census tables establish the task universe; the run records and the coverage check, not the tables, certify the negative results.

What the two incumbents show. Theorems 2 and 3 make precise where a fixed-modulus exchange search is blind. Within the family $\lambda(n) \mid M$ the search is exhaustive, so no choice of base, radius or insertion pool at M or at any divisor of M can reach a number outside it; Webster's N lies outside the family of N_0 by exactly one prime power (263, through 1579), and no search at M_0 could have found it. For the two families searched here the lever for further improvement is therefore the modulus rather than the radius (in other, incompletely searched families a larger radius can still matter). Once a candidate λ is proposed, the census and the search of Lemma 3 either produce a smaller number or certify that none exists in that family; for the two families here this took minutes per modulus, an observation about these computations rather than a bound for arbitrary moduli, whose pools and feasible radii may be much larger. Webster's number supplied a second modulus to investigate. Precisely what the two restricted minima establish is

$$n < N, n \text{ a 64-factor Carmichael number} \implies \lambda(n) \nmid M_0 \text{ and } \lambda(n) \nmid \lambda(N).$$

This leaves considerable uncertainty about both the size and the Carmichael function of S_{64} : the certified lower bound of §3 is 10^{145} (with 10^{146} at the tier of the strengthening remark in §4), and N exceeds 10^{146} by a factor of about 52.

3 Lower bound

Fix $Y = 10^{145}$. We show no 64-factor Carmichael number lies below Y .

3.1 A certified prime universe

Lemma 4 (Cofactor lower bound). *There is an explicit constant*

$$\begin{aligned} C = & 181214059509297180446495905961257618839971352 \\ & 500899762007368983685095561308686715879985893 \\ & 492155734750437055024004408677366582371670140 \\ & 3213549. \end{aligned}$$

(a 142-digit integer) such that for every Carmichael number n with $\omega(n) = 64$ and every prime factor $q \mid n$, the product of the remaining 63 factors satisfies $n/q \geq C$.

Proof (finite computation). Fix such an n and a prime factor q ; let $p_1 < \dots < p_{63}$ be the other factors, all odd (Lemma 1) and pairwise admissible. Let S be the set of these that are ≤ 97 ; there are finitely many internally admissible such S . For a fixed S , let $b_1(S) < b_2(S) < \dots$ be the primes > 97 that are *individually* admissible with every element of S (i.e. $p \nmid (b-1)$ for all $p \in S$). The actual cofactor tail $r_1 < \dots < r_{63-|S|}$ of n consists of primes > 97 each admissible with S , so $r_i \geq b_i(S)$ for every i . Hence

$$n/q = \prod_{p \in S} p \cdot \prod_{i=1}^{63-|S|} r_i \geq F(S) := \left(\prod_{p \in S} p \right) \prod_{i=1}^{63-|S|} b_i(S) \geq \min_S F(S) = C.$$

Three points make this rigorous: (i) the $b_i(S)$ need not be pairwise admissible with one another, but ignoring those further constraints only decreases the product, so $F(S)$ is a genuine lower bound; (ii) the computation uses a finite prime table (odd primes $\leq 10^4$) and checks, for every enumerated S , that it contains enough eligible tail primes, so no smaller completion is truncated away; (iii) the cofactor n/q itself need not be Carmichael. The minimum is taken over all 407760 admissible S . Thus C is a certified lower bound on the cofactor product; it is not claimed to be the smallest possible admissible 63-factor product. $\square$

Corollary 1 (Universe). *Every 64-factor Carmichael number $n < Y$ has largest prime factor*

$$P^+(n) = q \leq \left\lfloor \frac{Y-1}{C} \right\rfloor = 5518.$$

In particular every prime factor of such an n lies among the 727 odd primes ≤ 5518 (the largest being 5507).

Proof. Take $q = P^+(n)$ the largest factor; then $n = q \cdot (n/q) \geq qC$ by Lemma 4, and $n < Y$ gives $q \leq (Y-1)/C$; evaluate the floor. Every other factor is smaller, hence also ≤ 5518 . $\square$

3.2 Exhaustion

Theorem 4 (Lower bound). *There is no Carmichael number $n < 10^{145}$ with $\omega(n) = 64$. Hence $S_{64} \geq 10^{145}$.*

Proof (finite exhaustion). By Corollary 1 every candidate is a product of 64 distinct primes drawn from the universe U of 727 odd primes ≤ 5518 , with product $< Y$. Enumerate all increasing 64-subsets of U with product $< Y$, pruning a partial choice only by two *necessary* conditions: (i) fewer than the required number of primes remain; and (ii) the current product times the 64 – h smallest still-available primes individually compatible with the h already chosen already reaches Y (an exact-integer $R_{\min}$ test). Pairwise admissibility (Lemma 1) is enforced during descent—once p is chosen, every later prime q with $p \mid (q - 1)$ is discarded—which removes no genuine Carmichael candidate, since any real completion survives pairwise filtering and its remaining factors lie in the current available pool. At each complete 64-product, test Korselt.

The enumeration terminates with exactly 127092 increasing pairwise-admissible 64-subsets of U whose product is below Y (equivalently 127092 distinct products, by unique factorization), of which *none* is Carmichael. Therefore no 64-factor Carmichael number is smaller than 10^{145} . $\square$

3.3 Reproducibility

The exhaustion was carried out by *three separate enumeration implementations*, and the universe bound C was *independently recomputed*; all agree. Everything below is at repository revision f872e32 of [AlexanderEastwood/carmichael-frontier](https://github.com/AlexanderEastwood/carmichael-frontier).

1. `s64_lower_bound_certificate.py` runs two enumerators: a binary include/exclude walk with an incrementally maintained $\text{lcm}(p_i - 1)$ Korselt test (327049 reference nodes), and an ordered-choice walk using per-prime divisibility, a separate prime generator, and no bitmasks (163525 nodes). Both report 127092 admissible products and 0 Carmichael. They use different prime generators, traversal logic, and Korselt predicates, while *sharing* the cofactor-bound computation and the results/hash reporting code; the identical order-sensitive leaf digest `sha256 = e05f62598eea784b3f60f025c6ccbf30700115ecbef42702e24154dcece32321` is therefore a reproducibility aid, not an additional independent correctness check.
2. `carmichael_prime_bound.py` independently recomputes C and the cap $P^+ \leq 5518$ from first principles. It does *not* enumerate products or establish the absence of Carmichael numbers; it certifies only the universe.
3. `s64_lower_bound_oracle_check.py` is a separate clean-room enumerator (its own prime generation and ordered traversal) that takes the Korselt verdict at every one of the 127092 leaves from the frozen oracle `ref/ref_carmichael.py::verify_certificate` rather than from its own arithmetic, again finding 0 Carmichael. It was validated against the oracle's brute-force `least_with_k` on small k : $k = 3 \rightarrow 561$, $k = 4 \rightarrow 41041$, $k = 5 \rightarrow 825265$.

Exact commands:

```
python3 s64_lower_bound_certificate.py
python3 carmichael_prime_bound.py --X 1e145 --pi
python3 s64_lower_bound_oracle_check.py
```

4 The interval

Combining Theorem 4 (lower) with §2 (upper) proves Theorem 1: $10^{145} \leq S_{64} \leq N < 10^{148}$, so S_{64} has 146, 147, or 148 digits, with remaining exclusion interval $[10^{145}, N)$. This is a global statement about S_{64} ; it is *not* a determination of S_{64} , which would require excluding every 64-factor Carmichael number in $[10^{145}, N)$.

Remark 2 (Strengthening to 147–148 digits). A complete exact- k preproduct search over all 64-factor products $\leq 10^{146}$ returns no Carmichael number, giving $10^{146} < S_{64} \leq N$, i.e. S_{64} has 147 or 148 digits. This is a different tool from the enumerators of §3.3 (a depth-first preproduct search with an exact $R_{\min}$ bound and the $P\lambda(P) > B$ arithmetic-progression switch), and it is far cheaper than the $\geq 3.89 \times 10^{10}$ admissible-product count below 10^{146} because it rejects prefixes without materialising each product (≈ 71 s, single core). *Verification tier*: the $S_{64} \geq 10^{145}$ bound of Theorem 4 is the triple-verified core; this strengthening currently rests on the search engine, corroborated by two independent switch strategies both returning “none” and by the same engine reproducing “none” at 10^{145} (a fourth independent confirmation of that bound) and reproducing $S_{10} \dots S_{29}$ against OEIS A006931. An independent admissibility certificate at 10^{146} is the outstanding follow-up; the full exclusion of $[10^{145}, N)$ that would prove $S_{64} = N$ remains a machine-scale endgame.

5 Transfer to S_{65}

The cofactor machinery transfers to $k = 65$ with no new exhaustion. Let

$$A_{64} = \min\{\prod_{i=1}^{64} p_i : p_1 < \dots < p_{64} \text{ odd primes, pairwise admissible}\}$$

be the least product of 64 pairwise-admissible odd primes. The enumeration of §3 records it as a by-product: the proof of Lemma 4 never uses that the cofactor belongs to a Carmichael number, so it bounds every pairwise-admissible 63-product by C ; hence every admissible 64-product below 10^{145} has all its primes ≤ 5518 and is among the 127092 enumerated products, and the minimum over the enumeration is the global A_{64} (that this minimum lies below 10^{145} is witnessed by those products). It is the explicit 145-digit integer

$$\begin{aligned} A_{64} = & 441172365445881169056428109132100061485293342 \\ & 780359484110673067879903768586477795940804550 \\ & 206689631876408186963205931713339027583328239 \\ & 6201943377. \end{aligned}$$

Lemma 5 (Cofactor transfer). *For every Carmichael number n with $\omega(n) = 65$ and every prime $q \mid n$, the remaining 64 factors form a pairwise-admissible 64-product (Lemma 1), so $n/q \geq A_{64}$. Hence any such $n < U$ has $P^+(n) \leq \lfloor (U-1)/A_{64} \rfloor$.*

Taking $U = 10^{148}$ gives $P^+ \leq 2266$: a universe of 334 odd primes.

Theorem 5 (Lower bound for S_{65}). *There is no Carmichael number $n < 10^{148}$ with $\omega(n) = 65$. Hence $S_{65} \geq 10^{148}$, and the smallest 65-factor Carmichael number has at least 149 digits.*

Proof (finite exhaustion). By Lemma 5 every candidate is a product of 65 distinct primes from the 334-prime universe with product $< 10^{148}$. An ordered-prefix search with exact product pruning descends until $P\lambda(P) > 10^{148}$ and then tests the at most one cofactor in the forced residue class $R \equiv P^{-1} \pmod{\lambda(P)}$, enforcing 65 distinct factors and full Korselt with primality taken from the sieve. It visits 64355 nodes, forms 66 progression candidates, and finds no Carmichael number. $\square$

Remark 3 (Reproduction and verification tier). Theorem 5 was reproduced four ways, all agreeing on 0 Carmichael numbers: (i) the certificate `s65/s65_exclusion.py` above; (ii) a separate native binary enumeration `s65/verify65.cpp` that checks all 22,287,138 admissible 65-products below

10^{148} without inverses or the progression shortcut; and (iii)–(iv) our preproduct search engine in both switch modes (§3.3), which independently reports $n = \text{none}$ with matching counts (5012 inversions, 66 progression candidates). A companion certificate `s65/s65_transfer_certificate.py` establishes the weaker $S_{65} \geq 3.17 \times 10^{147}$ by extending the recorded 64-products two independent ways (285 admissible 65-products, 0 Carmichael). All rest on the same A_{64} , whose defining 127092-product set and digest are the triple-verified core of §3.

Theorem 6 (Interval for S_{65}). $10^{148} \leq S_{65} \leq U_{65}$, where U_{65} is the 151-digit Carmichael number with exactly 65 prime factors 19, 31, 37, $\dots$, 1021, 1993 (Webster’s candidate). Consequently S_{65} has 149, 150, or 151 decimal digits.

Proof. The lower bound is Theorem 5. For the upper bound: by Lemma 5 every 65-factor Carmichael below 10^{151} has all prime factors $\leq \lfloor (10^{151} - 1)/A_{64} \rfloor = 2266687$, so U_{65} (being $< 10^{151}$) factors completely over that range. Independent trial division confirms U_{65} is a product of exactly 65 distinct primes ≤ 1993 ; the frozen oracle `ref/ref_carmichael.py::verify_certificate` confirms Korselt, so U_{65} is Carmichael with $\omega = 65$ and $S_{65} \leq U_{65}$. As U_{65} has 151 digits and $S_{65} \geq 10^{148}$ has at least 149, the interval follows. (Factor list in `results_k65_webster_verified.json`.) $\square$

6 An interval ladder for $k = 64, \dots, 144$

Lemma 4 is an instance of a bound available at every factor count. For $m \geq 1$ and a cutoff y , let $\mathcal{H}_y$ be the internally admissible subsets of the odd primes $\leq y$, and for $S \in \mathcal{H}_y$ let $b_1(S) < b_2(S) < \dots$ be the primes $> y$ individually admissible with every element of S . Define

$$C_m(y) = \min_{\substack{S \in \mathcal{H}_y \\ |S| \leq m}} \left(\prod_{p \in S} p \right) \prod_{i=1}^{m-|S|} b_i(S),$$

and

$$A_m = \min \left\{ \prod_{i=1}^m p_i : p_1 < \dots < p_m \text{ odd, pairwise admissible} \right\}.$$

Theorem 7 (Ladder). For every $m \geq 3$ for which S_m exists, $C_m(y) \leq A_m \leq S_m$. Hence, for each k with a verified incumbent U_k , $C_k(y) \leq S_k \leq U_k$.

Proof. $A_m \leq S_m$ because every Carmichael number is squarefree with pairwise admissible prime factors (Lemma 1), so a k -factor Carmichael number is in particular an admissible k -product. $C_m(y) \leq A_m$ is the argument of Lemma 4 with 63 replaced by m : for an admissible m -product write S for its primes $\leq y$; its tail primes exceed y and are each admissible with S , so the i -th smallest is $\geq b_i(S)$, and ignoring the tail’s mutual constraints only lowers the product. Coverage requires enough eligible tail primes for every class, which is checked. $\square$

The three quantities are deliberately kept distinct: $C_m(y)$ is a certified relaxed bound (what the table uses), A_m the exact admissible minimum (a separate optimisation, not needed here and not certifiable by the Carmichael residue shortcut, since an admissible product need not satisfy $n \equiv 1 \pmod{\lambda(n)}$), and S_m the Carmichael minimum. A completed Carmichael exclusion below Y strengthens a row’s lower endpoint; it is *not* a bound on admissible products and must not be substituted into a cofactor formula.

All $C_m(97)$ for $63 \leq m \leq 144$ were obtained in one shared pass over the 407760 classes of $\mathcal{H}_{97}$ (tail primes $\leq 10^4$, the largest actually needed being 3209; exact integer arithmetic, ≈ 10 s).

$C_{63}(97)$ reproduces the constant of Lemma 4, and $C_{64}, C_{100}, C_{137}, C_{144}$ agree with an independent computation. The upper bounds U_k are verified k -factor Carmichael numbers: $U_{64} = N$ and U_{65} are Webster’s (§2, §5), the rows $k = 66, \dots, 144$ come from our search; each is re-checked by the frozen oracle. Table 1 lists the resulting digit intervals; the rows $k = 64, 65$ carry the stronger completed-exclusion endpoints of §3–§5. The elementary primorial baseline is shown for comparison; the relaxed bound improves it by 16–29 digits across the ladder (column “gain”). This is the improvement we can identify and quantify: the product of the first k odd primes is the elementary lower bound available without the admissibility argument, and for $k \geq 64$ we know of no published lower bound on S_k to compare against. The *exact* endpoints—the full integers $C_k(97)$, the incumbents U_k with factor lists, sources, and verification flags—are the certificate and are recorded per row in `ladder/ladder_manifest.json`, generated by `ladder/gen_ladder.py`; the table below is rendered from that manifest by `ladder/gen_ladder_tex.py`, so no value is hand-transcribed.

Table 1: Certified digit bounds $C_k(97) \leq A_k \leq S_k \leq U_k$, $k = 64, \dots, 144$. “primorial” = digits of the product of the first k odd primes (the elementary baseline); “gain” = digits by which $C_k(97)$ exceeds it; “lower endpoint” = the certified lower bound actually used (a completed Carmichael exclusion where one exists, else $C_k(97)$); every U_k re-verified by the frozen oracle. Exact endpoints are in `ladder/ladder_manifest.json`.

k	primorial	$C_k(97)$	gain	lower endpoint	digits of S_k	U_k	$P^+(U_k)$
64	128	144	+16	none $\leq 10^{146}$	[147, 148]	148	1579
65	130	147	+17	none $< 10^{148}$	[149, 151]	151	1993
66	133	150	+17	$C_{66}(97)$	[150, 154]	154	3221
67	136	153	+17	$C_{67}(97)$	[153, 158]	158	2311
68	138	155	+17	$C_{68}(97)$	[155, 161]	161	2393
69	141	158	+17	$C_{69}(97)$	[158, 164]	164	2347
70	143	161	+18	$C_{70}(97)$	[161, 166]	166	2971
71	146	163	+17	$C_{71}(97)$	[163, 168]	168	1471
72	148	166	+18	$C_{72}(97)$	[166, 173]	173	2971
73	151	169	+18	$C_{73}(97)$	[169, 176]	176	2731
74	153	172	+19	$C_{74}(97)$	[172, 178]	178	2731
75	156	174	+18	$C_{75}(97)$	[174, 183]	183	3221
76	159	177	+18	$C_{76}(97)$	[177, 185]	185	3169
77	161	180	+19	$C_{77}(97)$	[180, 187]	187	3121
78	164	183	+19	$C_{78}(97)$	[183, 191]	191	3911
79	166	186	+20	$C_{79}(97)$	[186, 194]	194	3823
80	169	188	+19	$C_{80}(97)$	[188, 197]	197	3727
81	172	191	+19	$C_{81}(97)$	[191, 200]	200	3361
82	174	194	+20	$C_{82}(97)$	[194, 203]	203	2971
83	177	197	+20	$C_{83}(97)$	[197, 205]	205	2731
84	180	199	+19	$C_{84}(97)$	[199, 209]	209	3313
85	182	202	+20	$C_{85}(97)$	[202, 211]	211	3529
86	185	205	+20	$C_{86}(97)$	[205, 215]	215	3911
87	188	208	+20	$C_{87}(97)$	[208, 219]	219	2381
88	190	211	+21	$C_{88}(97)$	[211, 221]	221	3329
89	193	214	+21	$C_{89}(97)$	[214, 223]	223	3727
90	196	216	+20	$C_{90}(97)$	[216, 227]	227	3911
91	198	219	+21	$C_{91}(97)$	[219, 230]	230	3037
92	201	222	+21	$C_{92}(97)$	[222, 234]	234	3137
93	204	225	+21	$C_{93}(97)$	[225, 236]	236	3301

k	primorial	$C_k(97)$	gain	lower endpoint	digits of S_k	U_k	$P^+(U_k)$
94	206	228	+22	$C_{94}(97)$	[228, 239]	239	3727
95	209	231	+22	$C_{95}(97)$	[231, 242]	242	2801
96	212	233	+21	$C_{96}(97)$	[233, 245]	245	3433
97	214	236	+22	$C_{97}(97)$	[236, 248]	248	3697
98	217	239	+22	$C_{98}(97)$	[239, 252]	252	2731
99	220	242	+22	$C_{99}(97)$	[242, 255]	255	3571
100	223	245	+22	$C_{100}(97)$	[245, 257]	257	3529
101	225	248	+23	$C_{101}(97)$	[248, 261]	261	2017
102	228	251	+23	$C_{102}(97)$	[251, 264]	264	3823
103	231	254	+23	$C_{103}(97)$	[254, 268]	268	3529
104	234	256	+22	$C_{104}(97)$	[256, 270]	270	3433
105	236	259	+23	$C_{105}(97)$	[259, 273]	273	3221
106	239	262	+23	$C_{106}(97)$	[262, 276]	276	3823
107	242	265	+23	$C_{107}(97)$	[265, 280]	280	3329
108	245	268	+23	$C_{108}(97)$	[268, 283]	283	3511
109	247	271	+24	$C_{109}(97)$	[271, 287]	287	3137
110	250	274	+24	$C_{110}(97)$	[274, 290]	290	3851
111	253	277	+24	$C_{111}(97)$	[277, 293]	293	3673
112	256	280	+24	$C_{112}(97)$	[280, 297]	297	3169
113	259	283	+24	$C_{113}(97)$	[283, 300]	300	3361
114	261	286	+25	$C_{114}(97)$	[286, 304]	304	3727
115	264	289	+25	$C_{115}(97)$	[289, 307]	307	3329
116	267	291	+24	$C_{116}(97)$	[291, 310]	310	3851
117	270	294	+24	$C_{117}(97)$	[294, 314]	314	3361
118	273	297	+24	$C_{118}(97)$	[297, 318]	318	3529
119	275	300	+25	$C_{119}(97)$	[300, 320]	320	3823
120	278	303	+25	$C_{120}(97)$	[303, 324]	324	3571
121	281	306	+25	$C_{121}(97)$	[306, 326]	326	3851
122	284	309	+25	$C_{122}(97)$	[309, 333]	333	3673
123	287	312	+25	$C_{123}(97)$	[312, 336]	336	3361
124	290	315	+25	$C_{124}(97)$	[315, 337]	337	3529
125	292	318	+26	$C_{125}(97)$	[318, 341]	341	3529
126	295	321	+26	$C_{126}(97)$	[321, 344]	344	3697
127	298	324	+26	$C_{127}(97)$	[324, 350]	350	3851
128	301	327	+26	$C_{128}(97)$	[327, 351]	351	3329
129	304	330	+26	$C_{129}(97)$	[330, 355]	355	3673
130	307	333	+26	$C_{130}(97)$	[333, 359]	359	3673
131	310	336	+26	$C_{131}(97)$	[336, 361]	361	3851
132	313	339	+26	$C_{132}(97)$	[339, 363]	363	3697
133	315	342	+27	$C_{133}(97)$	[342, 369]	369	3851
134	318	345	+27	$C_{134}(97)$	[345, 374]	374	3911
135	321	348	+27	$C_{135}(97)$	[348, 381]	381	3851
136	324	351	+27	$C_{136}(97)$	[351, 380]	380	3673
137	327	354	+27	$C_{137}(97)$	[354, 386]	386	3911
138	330	357	+27	$C_{138}(97)$	[357, 386]	386	3851
139	333	360	+27	$C_{139}(97)$	[360, 388]	388	3911
140	336	363	+27	$C_{140}(97)$	[363, 396]	396	3911
141	339	366	+27	$C_{141}(97)$	[366, 401]	401	3911
142	342	369	+27	$C_{142}(97)$	[369, 401]	401	3911
143	344	373	+29	$C_{143}(97)$	[373, 405]	405	3911
144	347	376	+29	$C_{144}(97)$	[376, 406]	406	3911

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The known values for $3 \leq k \leq 63$ are due to R. G. E. Pinch (earlier k) and, for $k = 36$ – 63 , the Butler University computation; both are collected in J. Webster’s [small-carmichael-numbers](#), the repository cited here for the known values and the N_{63} , N_{65} neighbours. Two existence results, of different scope, are relevant and should not be conflated: Alford, Grantham, Hayman, and Shallue constructed Carmichael numbers with exactly k prime factors for every $3 \leq k \leq 19,565,220$; Larsen and Wright proved that Carmichael numbers exist for every sufficiently large factor count. Neither determines a minimum; the contribution reported here is an explicit interval for the minimum S_{64} .

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